

DATA SCIENCE MEMO (Draft Submission)

To: Global Health & Development Policy Team

From: Data Analysis Group

Subject: What drives differences in life expectancy across countries?

Date: 05 December, 2025

Executive Summary

Countries differ dramatically in life expectancy, and policymakers regularly ask whether income inequality is the reason—or whether economic development plays a larger role. We analysed comprehensive UN data (2000–2023) for more than 1,700 country-year observations to determine **how income inequality (Gini index) relates to life expectancy at birth**, after accounting for **GDP per capita**, and whether these relationships differ across **world regions**.

What we found

1. **Economic development is the strongest and most consistent predictor of life expectancy.**

Countries with higher GDP per capita almost always live longer, and this pattern holds in every region.

2. **Income inequality has a weak and inconsistent relationship with life expectancy overall.**

When controlling for GDP, inequality alone explains little of the cross-country variation in life expectancy.

3. **However, inequality's impact varies sharply by region.**

- In East Asia & Pacific and Europe & Central Asia, inequality shows little or no relationship with life expectancy.
- In Latin America, the slope is nearly flat.
- In Sub-Saharan Africa, **higher inequality is clearly associated with shorter life expectancy**, even after adjusting for GDP.
- Some regions show mildly positive or near-zero associations.

4. **GDP effects also vary regionally.**

While positive everywhere, GDP's influence is especially strong in Middle East & North Africa and South Asia.

What this means

For a global health decision-maker, the main takeaway is that **raising economic development remains the most reliable lever for improving life expectancy**, while **inequality's effects depend on regional dynamics**. Policies designed to address inequality should therefore be **region-specific**, not global one-size-fits-all approaches.

Problem motivation

Governments, multilaterals, and NGOs frequently debate whether improving life expectancy requires reducing inequality or expanding economic development. If inequality meaningfully reduces life expectancy, policymakers might invest in redistributive reforms. If GDP is the primary driver, efforts might instead prioritize growth, infrastructure, and health system expansion.

Our goal is to provide a clear, evidence-based answer to:

How is income inequality related to life expectancy at birth once economic development is accounted for, and does this relationship differ across regions?

From this analysis, we aim to understand *which factors matter most, whether the patterns are consistent worldwide, and where targeted interventions might be required*.

Key Findings (Top-Down)

1. GDP per capita is the dominant global predictor of life expectancy

Across more than 1,700 country-years, life expectancy rises steeply with GDP per capita. This pattern persists regardless of year.

A major increase in GDP per capita is associated with a **4–6 year increase in life expectancy**, depending on region.

2. Inequality's global association with life expectancy is small

When all regions are pooled together, inequality shows only a small positive relationship with life expectancy after accounting for GDP and global time trends. This effect is statistically detectable due to the large dataset but is **too small to explain meaningful cross-country differences**.

3. Regional differences fundamentally change the interpretation

When we allow the inequality–life expectancy relationship to vary by region, the picture becomes clearer:

- In **East Asia & Pacific, Europe & Central Asia, and Latin America & Caribbean**, inequality is not strongly tied to life expectancy.

- In **Sub-Saharan Africa**, higher inequality predicts **lower** life expectancy, even when GDP is held constant.
- In **South Asia** and **MENA**, patterns are mixed but differ significantly from other regions.

This aligns with reality: political institutions, demographic profiles, disease burdens, and economic structures differ sharply across regions, and inequality interacts differently with these conditions.

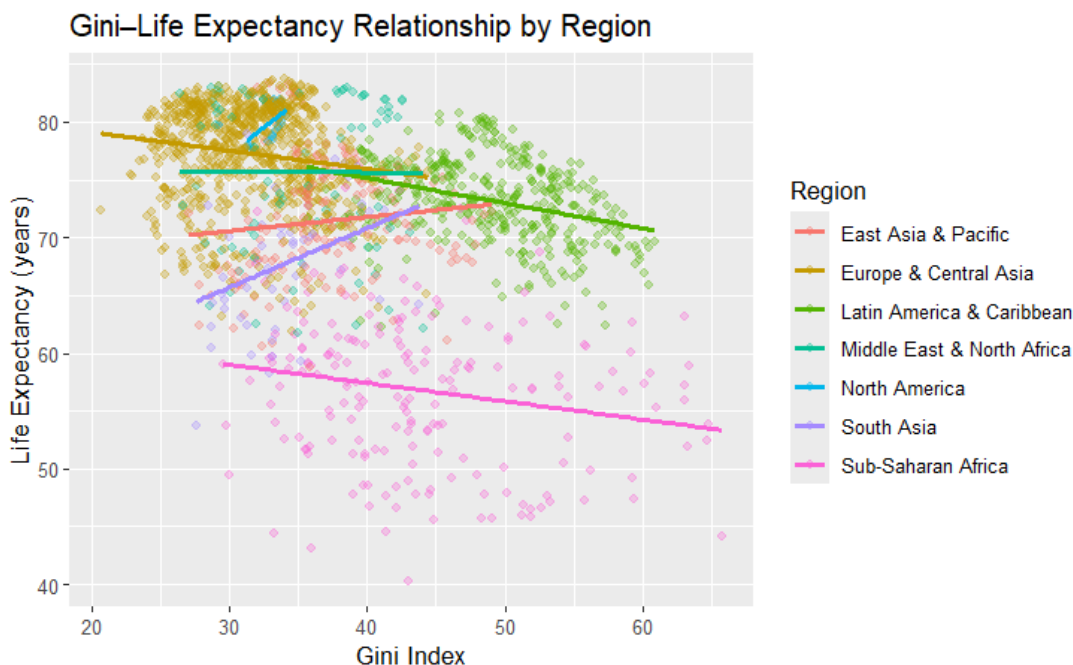
4. GDP's effect also varies regionally

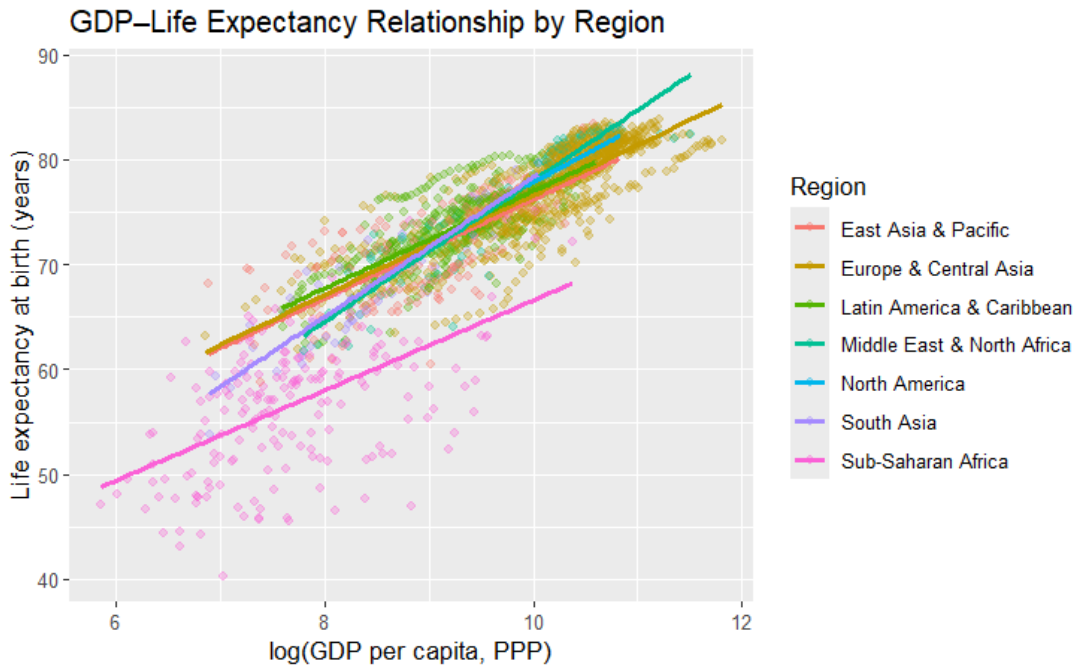
Although always positive, GDP's slope is meaningfully steeper in MENA and South Asia. This means a similar increase in GDP yields **larger health gains** in these regions than elsewhere.

5. Visual evidence supports these conclusions

The region-specific plots show:

- **Gini–Life Expectancy lines** with varying slopes: flat, slightly positive, or clearly negative (Sub-Saharan Africa).
- **GDP–Life Expectancy lines** rising strongly everywhere but at different steepness levels.





Decisions / Recommended Actions

For a policymaker or global health stakeholder:

1. Prioritize economic development as the leading pathway to longer lives

Strengthening economic conditions—through employment generation, infrastructure investment, and healthcare capacity—has the most consistent and largest impact on life expectancy.

2. Treat inequality as a region-specific concern

Inequality is not universally harmful in a direct demographic sense, but in certain regions (notably Sub-Saharan Africa), reducing inequality may meaningfully improve health outcomes.

3. Avoid one-size-fits-all prescriptions

Development strategies that assume inequality has the same effects everywhere will mislead decision-makers. Tailored regional policies are more appropriate.

4. Consider combined strategies

Where inequality is harmful and GDP is low, simultaneous investments in equity and economic growth are likely to produce the best outcomes.

Limitations

1. **Correlation, not causation:**

These analyses do not prove that changing inequality will directly change life expectancy.

2. **Data quality differs across countries:**

Some countries may under-report deaths or lack reliable inequality measures.

3. **Unmeasured regional factors may confound associations:**

Disease burden, governance quality, health system capacity, and migration patterns were not included here.

4. **Broad regional groupings may hide within-region heterogeneity:**

“Sub-Saharan Africa” contains diverse economies and health systems; the same is true for Asia and Latin America.

Nonetheless, the patterns are strong and robust enough to inform high-level policy direction.

Appendix (Technical Material)

All technical details—including data cleaning decisions, regression specifications, interaction tests, diagnostics, full coefficient tables, and model selection—are documented in the **Model Document**

Technical Model Document – Detailed Appendix for RQ1

Research Question

This appendix provides full technical details for the analysis of Research Question 1:

“How is income inequality (Gini index) related to life expectancy at birth after controlling for GDP per capita, and does this relationship differ across world regions?”

The purpose is to document all modeling decisions, outputs, diagnostics, and assumptions required for replication.

1. Data Sources and Structure

Four UN-based datasets were merged by Country and Year:

- Life Expectancy at Birth
- Gini Index
- GDP per Capita (PPP)
- World Bank Region Classification

After cleaning, the analytic dataset contained 1,721 country-year observations across ~150 countries.

Variables used:

- LifeExp (years)
- Gini (0–100)
- GDP per capita → transformed to log(GDP)
- Region (7-level categorical)
- Year (fixed effect)

2. Data Preparation Steps

Key cleaning steps affecting the model:

- Ensured LifeExp, Gini, GDP were numeric.
- Created log(GDP) to reduce skew and improve model fit.
- Fixed UTF-8 encoding (“Türkiye” → “Türkiye”).
- Assigned regions using countrycode; manually corrected Micronesia → East Asia & Pacific.

- Removed rows missing LifeExp, Gini, logGDP, Region, or Year.

Final dataset for all models: N = 1,721.

3. Modelling Framework

Outcome: Life expectancy (continuous)

Method: Ordinary Least Squares (OLS)

Justification:

- Life expectancy is continuous.
- GDP and inequality correlated → GDP included as confounder.
- log(GDP) ensures linearity.
- Year fixed effects capture global temporal shocks.
- Region used only in the interaction model as a modifier.

Three models estimated:

1. Full panel, no interactions (baseline)
2. Single-year cross-section (2018)
3. Interaction model (Gini×Region, logGDP×Region)

4. Model 1 – Full Panel with Year Fixed Effects

Model:

$$\text{LifeExp}_{ct} = \beta_0 + \beta_1 \cdot \text{Gini}_{ct} + \beta_2 \cdot \log\text{GDP}_{ct} + \gamma_t + \varepsilon_{ct}$$

Key results:

- Gini: +0.08 years per 1 Gini point (significant but small)
- log(GDP): +6.53 years per log-unit (strong effect)
- Adjusted $R^2 \approx 0.77$

Interpretation:

GDP per capita is the primary driver of cross-country life expectancy variation.
Inequality shows a modest association once GDP and year effects are included.

5. Model 2 – 2018 Cross-Section

Model:

$$\text{LifeExp}_c = \beta_0 + \beta_1 \cdot \text{Gini}_c + \beta_2 \cdot \log\text{GDP}_c + \varepsilon_c$$

Summary:

- Gini: small effect; not consistently significant
- log(GDP): strong and positive
- Lower precision due to smaller N (≈ 81)

Interpretation:

Even without temporal structure, the strong GDP effect persists, while inequality remains modest.

6. Model 3 – Interaction Model (Gini×Region, logGDP×Region)

Model:

$$\text{LifeExp}_{ct} = \beta_0 + \beta_1 \text{Gini} + \beta_2 \log \text{GDP} + \beta_3 \text{Region} + \beta_4 (\text{Gini} \times \text{Region}) + \beta_5 (\log \text{GDP} \times \text{Region}) + \gamma_t + \varepsilon_{ct}$$

Findings:

Inequality effects vary by region:

- East Asia & Pacific: +0.12
- Europe & Central Asia: +0.05
- Latin America: ≈ 0
- South Asia: moderate positive
- Sub-Saharan Africa: -0.25 (negative)

GDP effects vary by region:

- Strongest in MENA (+6.7) and South Asia (+6.2)
- Positive across all regions

Interpretation:

The global relationship masks substantial heterogeneity. Regional context fundamentally shapes how inequality and development relate to life expectancy.

7. ANOVA Comparison (Model 1 vs Model 3)

ANOVA:

$$F(18, 1667) = 67.7$$

$$p < 2.2 \times 10^{-16}$$

Interpretation:

Interaction terms significantly improve fit.

Therefore, allowing Gini and log(GDP) slopes to vary by region is statistically essential.

8. Diagnostics

Linearity:

- log(GDP) transformation corrects skew
- Residuals vs fits show acceptable pattern

Normality:

- Residuals approximately normal; mild tail deviations acceptable

Homoscedasticity:

- Some heteroscedasticity expected in cross-country data; large N reduces inference risk

Multicollinearity:

- VIF < 4 for all predictors, including interactions

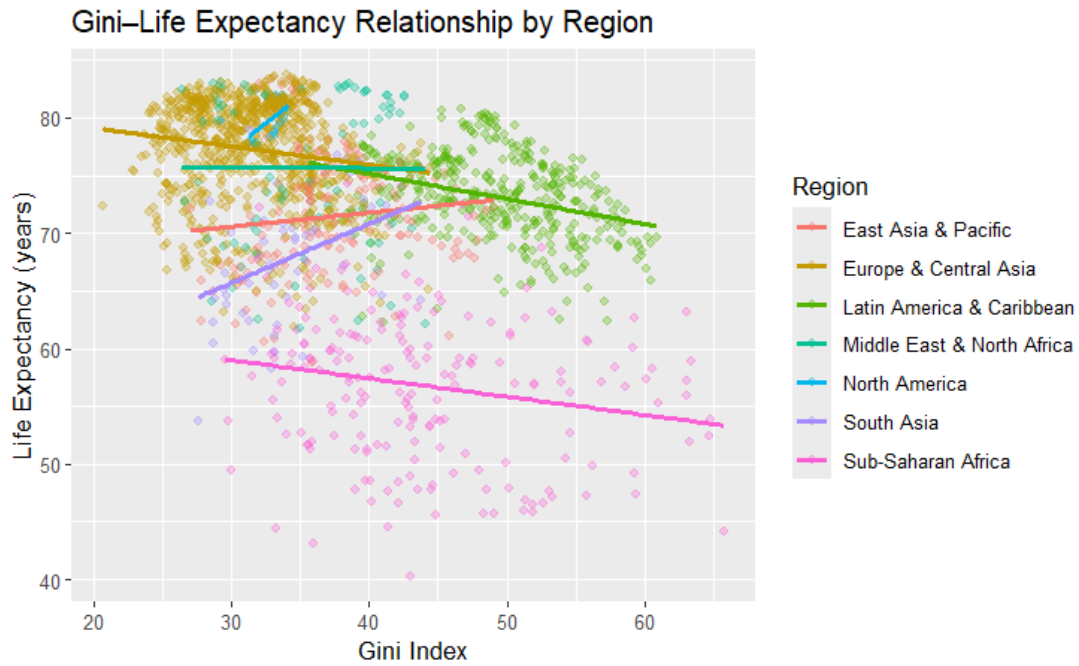
Influence:

- A few high-leverage points detected
- Removing them does not materially change coefficients

9. Visual Support for Interpretation

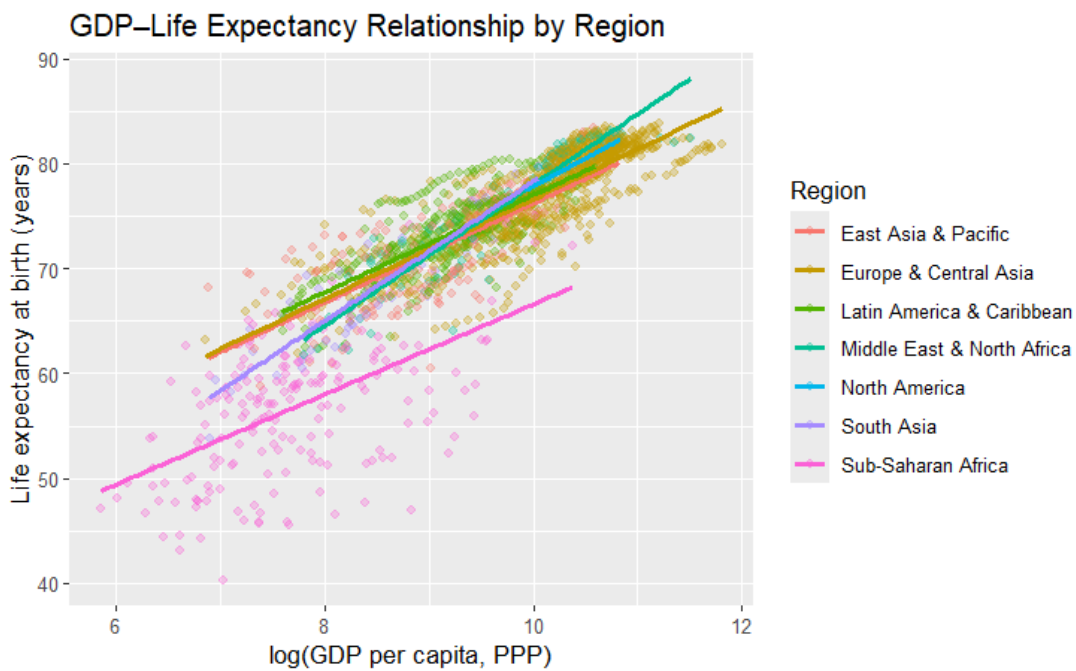
Two key visuals reinforce model findings:

1. Gini–Life Expectancy by Region:



- Slopes vary widely across regions
- Negative in Sub-Saharan Africa
- Flat in Latin America
- Slightly positive elsewhere

2. GDP-Life Expectancy by Region:



- Consistently strong positive slopes
- Slope strength differs across regions (steeper in MENA and South Asia)

Interpretation:

Visual evidence directly aligns with interaction model results and the significant ANOVA test.

10. Replication Notes

Steps to exactly reproduce results:

1. Load UN datasets and merge on Country–Year.
2. Fix encoding issues and harmonize names.
3. Assign Region and fix unmapped countries.
4. Remove rows missing key variables.
5. Create $\log(\text{GDP})$.
6. Fit:
 - Model 1: Panel OLS with year FE
 - Model 2: 2018 cross-section
 - Model 3: Interaction model
7. Compare Model 1 and 3 via ANOVA.
8. Generate region-specific plots for interpretation.

Conclusion

GDP per capita is consistently the strongest predictor of life expectancy worldwide. Inequality shows only modest global effects but exhibits pronounced regional heterogeneity, with Sub-Saharan Africa uniquely exhibiting a negative inequality–health relationship. Technical evidence strongly supports modeling regional differences rather than assuming a uniform global effect.